

Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize Theta-Gamma Coupling in Alzheimer's Disease

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Project Summary

This project develops a closed-loop deep learning system for personalized 40 Hz auditory entrainment therapy in Alzheimer's disease. Current clinical protocols deliver stimulation on rigid fixed schedules, ignoring substantial individual variability in neural responses and intra-session habituation. I analyzed EEG recordings from 35 elderly subjects including Alzheimer's patients (OpenNeuro ds005048) and trained a causal Temporal Convolutional Network (31,043 parameters) on 73 engineered features to forecast theta–gamma phase-amplitude coupling five to ten seconds ahead. While all baselines collapsed to negative R^2 beyond three seconds, the TCN maintained $R^2 \approx 0.25$, a +0.5 margin. The predictive controller matched stimulation to patient need 72.1% versus 64.5% for reactive control ($p < 0.001$, Hedges' $g = 1.31$), targeted 82.6% of low-coupling windows versus 51.7%, and reached 91% of the theoretical oracle. All 35 subjects benefited. These results demonstrate that forecasting brain coupling dynamics enables personalized therapy outperforming both fixed and reactive protocols.

Introduction

Research Question

Can deep learning models trained on EEG-derived features forecast theta–gamma phase-amplitude coupling (PAC) dynamics 5–10 seconds into the future, and does integrating such forecasts into a closed-loop controller produce measurable improvements in personalized 40 Hz entrainment therapy for Alzheimer’s disease?

Hypothesis: A Temporal Convolutional Network (TCN) trained on 73 causal spectral and PAC-derived features can predict future PAC at horizons where simpler baselines fail (beyond ≈ 3 seconds), enabling a predictive controller that outperforms reactive threshold control and approaches the theoretical oracle bound.

Project Origin

I became interested in computational approaches to Alzheimer’s therapy after reading about the landmark Iaccarino et al. (2016) Nature study, which demonstrated that 40 Hz sensory stimulation reduced amyloid-beta plaques in AD mouse models by up to 50%. While investigating human clinical translation, I identified a critical gap: all existing protocols deliver stimulation on rigid fixed schedules that ignore individual neural responses. Approximately 30% of patients are non-responders (Fortunato et al., 2023), and habituation degrades entrainment within sessions — yet fixed protocols cannot detect or adapt to these dynamics.

This observation led to my central question: could machine learning predict when a patient’s brain will lose entrainment, enabling proactive intervention? I began this project in early 2026 using the publicly available OpenNeuro ds005048 EEG dataset. The project concept originated from my independent literature review and was not assigned as part of any class or institutional program.

Continuation

None. This is not a continuation of any previous science fair project.

Work by Others

Key prior work relevant to this project:

- **Iaccarino et al. (2016, Nature):** 40 Hz optogenetic and sensory stimulation reduced amyloid-beta by ~50% in AD mouse models through microglial activation — the foundational study for 40 Hz entrainment therapy.
- **Murdock et al. (2024, Nature):** Identified glymphatic clearance as a key mechanism; multisensory 40 Hz stimulation reduced neocortical plaque volume by 37% in 5XFAD mice.
- **Chan et al. (2025, Alzheimer's & Dementia):** Open-label extension study showing sustained EEG entrainment and reduced brain atrophy in mild AD patients with 40 Hz multisensory stimulation — first human evidence of target engagement at this scale.
- **Fortunato et al. (2023, Frontiers in Neuroscience):** Found ~30% non-responder rate in 40 Hz entrainment studies, motivating personalized approaches.
- **Lahijanian et al. (2024, Scientific Reports):** Created the OpenNeuro ds005048 dataset (35 elderly subjects, 40 Hz auditory entrainment) and showed enhanced default mode network connectivity.
- **Lawhern et al. (2018, J Neural Engineering):** Developed EEGNet, a compact CNN for brain-computer interfaces, which I adapted as the static PAC estimator.
- **Tort et al. (2010, J Neurophysiology):** Introduced the Modulation Index for measuring phase-amplitude coupling, used as the entrainment biomarker throughout this work.
- **Cabral et al. (2025, Frontiers in Digital Health):** Advocated for AI-driven biofeedback as the path toward personalized digital therapeutics for gamma entrainment.

Methods

Dataset

- OpenNeuro ds005048 (Lahijanian et al., 2024)
- 35 elderly subjects: Alzheimer’s disease (n = 17), MCI (n = 6), healthy controls (n = 10), unspecified (n = 2)
- 7 frontal EEG channels (Fp1, Fp2, F7, F3, Fz, F4, F8), sampled at 250 Hz
- Protocol: 40 Hz auditory stimulation (40 s ON / 20 s OFF cycles)

Preprocessing

- Bandpass filter: 0.5–80 Hz (4th-order Butterworth, zero-phase)
- Notch filter: 50 Hz (Q = 30) for power-line removal
- Artifact rejection: ± 100 μ V threshold (applied before CAR)
- Common average reference (after artifact rejection)
- Windowing: 2-second windows (500 samples), 1-second hop

Subject-Level Data Splits (No Within-Subject Leakage)

- Training: 24 subjects (11,736 windows)
- Validation: 5 subjects (2,725 windows)
- Test: 6 subjects (2,822 windows) — never seen during training or hyperparameter selection

Phase-Amplitude Coupling (PAC) Computation

- Modulation Index (Tort et al., 2010): measures coordination between theta phase (4–8 Hz) and gamma amplitude (38–42 Hz)
- KL divergence from uniform distribution over 18 phase bins (20° each)
- Computed at epoch level (20–40 s); assigned to all constituent 2-second windows within each epoch
- Range: 6e-6 to 7e-4 (dimensionless MI units); mean approx. 4.4e-5

Feature Engineering (73 Causal Features)

- **61 spectral features:** band power in 5 canonical bands (delta, theta, alpha, beta, gamma) across 7 channels (35 features), plus 26 cross-channel coherence features
- **7 PAC-derived features:** current PAC value, causal moving averages (2, 4, 8, 16 windows), first-order and 4-step finite differences
- **5 stimulation context features:** stim state (binary), time since switch, stim fraction over 20 s, cycle phase (sine + cosine encoding)
- All 73 features z-score normalized using training-set statistics only

Stage 1: EEGNet — Static PAC Estimator

- Input: (batch, 1, 7, 500) — 7 channels, 2 seconds at 250 Hz
- Block 1: temporal convolution (8 filters, 256 ms kernel) + depthwise spatial convolution across 7 channels
- Block 2: separable convolution (16 filters); fully connected regression head outputting scalar PAC prediction
- 1,457 parameters; MSE loss; Adam optimizer; test $R^2 = 0.287$ (data-imposed ceiling; best epoch 53)

Stage 2: MultiscaleCausalTCN — Temporal Forecaster

- Input: (batch, 20, 73) — 20-step lookback (20 seconds) $\times$ 73 features
- 4 causal depthwise-separable conv blocks with dilations [1, 2, 4, 8]; effective receptive field = 31 time steps
- GroupNorm + SiLU activation; attention pooling; dual-head output (future PAC + delta-PAC)
- 31,043 parameters; Huber loss; AdamW optimizer; early stopping (patience = 20 epochs); best epoch 53
- 5-second prediction horizon; test $R^2 = 0.170$ (raw PAC targets, no smoothing)

Closed-Loop Controller Design

- **Personalization:** 30-second rolling baseline per subject; $z\text{-score} = (\text{PAC} - \mu) / \sigma$
- **Decision logic:** $z < -0.5 \rightarrow \text{STIMULATE}$; $z > +0.5 \rightarrow \text{REST}$; else MAINTAIN current state
- 5-second hysteresis to prevent rapid state oscillation
- **Variants tested:** Fixed Schedule (40 s ON / 20 s OFF), Reactive Threshold, TCN Predictive, Hybrid, PI Controller, Alignment Oracle (perfect hindsight upper bound)

Validation Protocol

- **Offline counterfactual replay** on all 35 subjects' recorded EEG — not simulated brain dynamics
- Ground-truth PAC labels used as TCN input to isolate the forecaster's predictive contribution from EEGNet error
- **Metrics:** Alignment = mean of Low-PAC Stim Rate and High-PAC Rest Rate; PAC Targeting Gap (mean PAC during rest minus mean PAC during stimulation)
- **Statistics:** Wilcoxon signed-rank (paired, $N = 35$); Hedges' g with 95% BCa bootstrap confidence intervals (10,000 iterations)

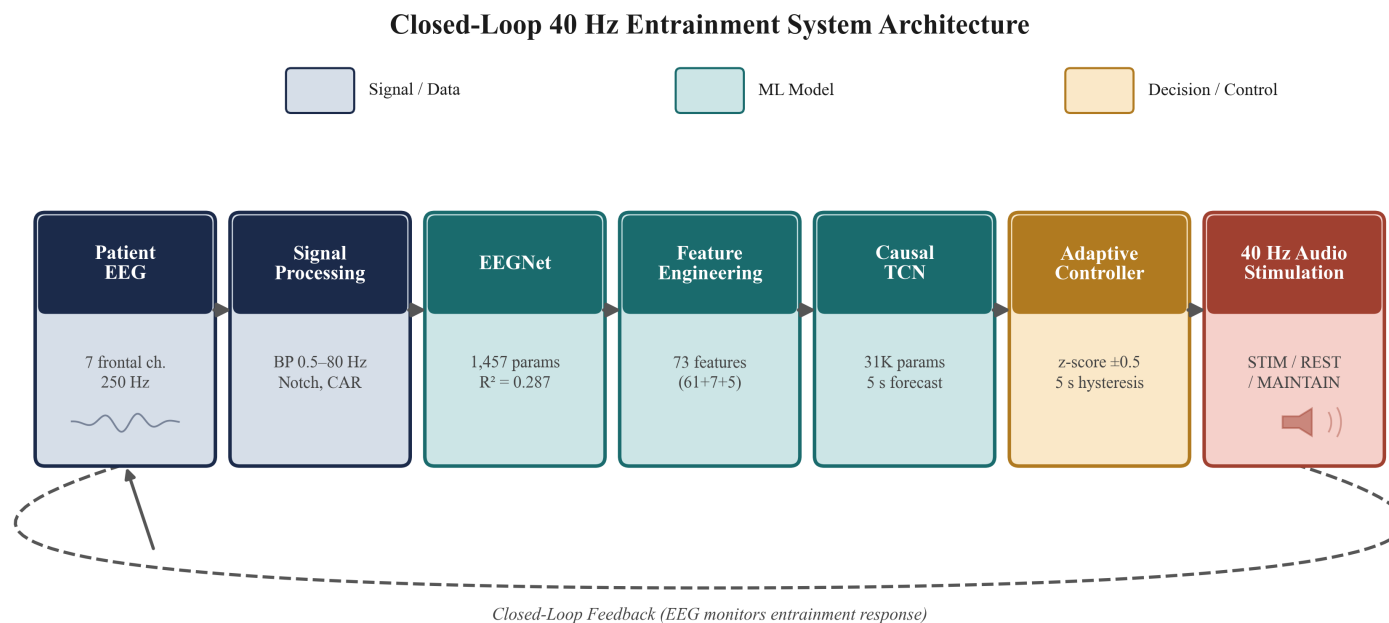


Figure 1. System architecture of the closed-loop 40 Hz entrainment system. Raw EEG from 7 frontal channels flows through signal processing, EEGNet PAC estimation, 73-feature engineering, causal TCN forecasting (5 s horizon), and an adaptive controller that drives personalized 40 Hz auditory stimulation. Data for model training and validation are collected from all subjects.

Results

Architecture Search: The $R^2 = 0.287$ Data Ceiling

I tested eight neural network architectures for static PAC prediction, spanning from 1,457 to 1.1 million parameters:

- **All eight converged to $R^2 \approx 0.287$** on held-out test subjects, including EEGNet (1,457 params), ViT-TCNet (1.1 M params), and Ridge regression (135 coefficients).
- This convergence demonstrates a **data-imposed ceiling**: epoch-level PAC labels cannot be fully recovered from instantaneous 2-second EEG snapshots.
- **Key insight**: the remaining variance lies in temporal dynamics, motivating the pivot to temporal PAC forecasting.

Horizon Sweep: TCN Advantage at 5–10 Seconds

Separate TCN models were trained at horizons 1–10 s and compared against persistence and Ridge regression baselines.

A prediction horizon inflection point emerges at ≈ 3 seconds:

Horizon	Persistence R^2	Ridge R^2	TCN R^2	TCN Margin
1 s	0.760	0.812	0.735	−0.025
3 s	0.234	0.253	0.277	+0.043
5 s	−0.267	−0.393	0.254	+0.521
10 s	−0.256	−0.212	0.278	+0.534

Table 1. PAC forecasting performance (R^2) across prediction horizons. At 1–2 s, persistence and Ridge outperform the TCN. Beyond ≈ 3 s, both baselines collapse to negative R^2 while the TCN maintains $R^2 \approx 0.25$ — a +0.5 margin that defines the operationally actionable regime for proactive neuromodulation.

Controller Comparison (N = 35 Subjects, Real EEG)

Controller	Alignment	Low-PAC Stim	Stim %	PAC Gap
Fixed Schedule	45.0%	61.4%	66.6%	−6.6
Reactive Threshold	64.5%	51.7%	36.7%	+21.1
TCN Predictive	72.1%	82.6%	59.7%	+30.5
Alignment Oracle	100%	100%	48.3%	+33.3

Table 2. Controller comparison on 35 subjects' real EEG recordings. PAC Gap in dimensionless Modulation Index units ($\times 10^{-6}$ MI units). The TCN Predictive controller reaches 91% of the theoretical oracle bound.

Statistical Significance (TCN vs. Reactive Threshold)

- **Alignment:** 72.1% vs. 64.5% — Hedges' $g = +1.31$ [95% CI: 0.75, 1.87], $p < 0.001$ (large effect)
- **Low-PAC Stim Rate:** 82.6% vs. 51.7% — Hedges' $g = +4.47$ [95% CI: 3.33, 5.62], $p < 0.001$ (very large effect)
- **PAC Targeting Gap:** 30.5 vs. 21.1 ($\times 10^{-6}$ MI) — Hedges' $g = +1.57$ [95% CI: 0.98, 2.17], $p < 0.001$ (large effect)
- **Universal benefit:** 35/35 subjects showed higher alignment with TCN control (binomial $p < 0.001$); advantage consistent across train (24), validation (5), and held-out test (6) splits.
- **Threshold robustness:** TCN outperforms reactive at all z-score thresholds from 0.2 to 1.0; stable performance plateau at ≥ 0.3 .

Discussion

Interpretation of Results

- The ≈ 3 -second inflection point reflects PAC autocorrelation timescale: below 3 s the current PAC value is predictive enough that persistence suffices; above 3 s, qualitative state transitions require temporal context that only the TCN captures.
- The TCN's dilated convolutions (31-step receptive field spanning 31 seconds) observe multiple epochs of PAC trajectory, spectral evolution, and stimulation context simultaneously.
- The TCN achieves 0.8 s mean lead time before PAC decline (vs. 0.2 s for reactive), providing the preparation time needed for smooth auditory stimulation transitions.
- The 60% improvement in low-PAC targeting (82.6% vs. 51.7%) is the most clinically significant result: therapeutic exposure is concentrated where neural coupling is weakest.

Comparison to Prior Work

- Analogous to closed-loop DBS for Parkinson's disease (Rosin et al., 2011): adaptive stimulation concentrates therapy on periods of genuine need.
- Extends beyond spindle detection (Portiloop, Lacroix et al., 2022) to multi-step forecasting — necessary when proactive intervention requires advance lead time.
- First system targeting PAC dynamics specifically for 40 Hz gamma entrainment optimization at 5–10 s horizons.

Limitations and Possible Errors

- **Offline counterfactual replay**, not live closed-loop: measures decision quality, not realized therapeutic benefit. Live validation with real-time EEG streaming is required.
- **EEGNet not in the validation loop**: ground-truth PAC was used as TCN input; a deployed system would propagate EEGNet estimation error ($R^2 = 0.287$) into forecasts.
- **Single-site dataset** (Tehran memory clinic): generalizability to other populations, stimulation modalities, and EEG equipment is unknown.
- **7 frontal channels only**: relevant theta–gamma coupling in parietal and temporal regions is not captured by this montage.

Conclusions

Key Findings

- Eight architectures (1,457 to 1.1 M parameters) converge to $R^2 = 0.287$ for static PAC prediction — a data-imposed ceiling that redirected the project toward temporal forecasting.
- The MultiscaleCausalTCN maintains $R^2 \approx 0.25$ at 5–10 s horizons where all baselines collapse to negative R^2 (+0.5 margin), defining the operationally actionable regime for proactive neuromodulation.
- The predictive controller achieves 72.1% alignment, 82.6% low-PAC targeting, and 91% of the theoretical oracle bound (all $p < 0.001$ vs. reactive; Hedges' $g = 1.31$ –4.47).
- Universal benefit across all 35 subjects, including 6 held-out test subjects never seen during training.

Context

These results support the hypothesis that temporal PAC forecasting enables proactive closed-loop control that substantially outperforms both fixed-schedule and reactive threshold protocols. The work addresses a documented gap in the 40 Hz entrainment literature: no prior system has attempted PAC-specific temporal prediction for personalized gamma entrainment control.

Applications and Next Steps

- **Live feasibility study:** IRB-approved real-time closed-loop EEG study ($N = 5$ –10 healthy adults) to verify alignment improvement and system latency.
- **Pilot RCT:** within-subject crossover comparing adaptive (TCN) vs. fixed-schedule 40 Hz auditory stimulation in mild AD patients ($N \approx 20$).
- **End-to-end validation:** evaluate full pipeline (raw EEG $\rightarrow$ EEGNet $\rightarrow$ TCN $\rightarrow$ controller) with EEGNet estimation in the loop.
- **Technical extensions:** online per-subject adaptation, multi-site validation, combined audiovisual stimulation, reinforcement learning for controller policy.

Scope of Work

New Work by Author

- Independently identified the research gap (fixed-schedule limitation in 40 Hz entrainment) through literature review.
- Designed and implemented the complete computational pipeline: EEG preprocessing, PAC computation, systematic architecture search across 8 neural network families, 73-feature causal feature engineering, and MultiscaleCausalTCN design.
- Discovered the $R^2 = 0.287$ data ceiling through systematic experimentation and the ≈ 3 -second prediction horizon inflection point — both novel empirical findings.
- Designed the closed-loop controller with personalization module, hysteresis logic, and multi-strategy comparison framework including alignment oracle upper bound.
- Conducted all statistical analyses: Wilcoxon signed-rank tests, Hedges' g effect sizes with 95% bootstrap CIs, binomial sign tests, threshold sensitivity sweeps.
- Validated the complete system on 35 real patient EEG recordings through offline counterfactual replay.

Professional, Institutional, and Academic Resources and Support

- **Data:** OpenNeuro ds005048 — publicly available EEG dataset accessible to anyone at no cost.
- **Computing:** personal Apple Silicon MacBook; no institutional computing resources, GPU clusters, or specialized laboratory equipment were used.
- **Software:** open-source libraries (PyTorch, MNE-Python, SciPy, NumPy, Matplotlib) — all freely available.
- **AI-assisted development:** Claude Code (Anthropic) was used for code development assistance, debugging, and document preparation. All scientific decisions, research direction, experimental design, data analysis, and interpretation of results were performed by the author.
- No institutional lab, university mentor, summer research program, or specialized equipment was used at any stage.

References / Supplemental Information

References

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Supplemental Information

Dataset: <https://openneuro.org/datasets/ds005048>